Linear Algebra and Calculus Lecture 8

Eric Sandin Vidal

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Matrix of a Linear Transformation

Theorem. *Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that $T(x) = Ax$ for all $x \in \mathbb{R}^n$. In fact, A is the $m \times n$ matrix whose j-th column is the vector $T(e_j)$, where e_j is the j-th column of the identity matrix in $\mathbb{R}^n$:*

$$A = [\underline{\quad\quad} \quad \underline{\quad} \quad \underline{\quad\quad}] .$$

Example. Let's find the standard matrix A for the dilation transformation $T(x) = 3x$, for x in $\mathbb{R}^2$. Here $T(e_1) = 3e_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ and $T(e_2) = 3e_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$. So $A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$.

Definition. A mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is said to be ____ $\mathbb{R}^m$ (or __________) if each b in $\mathbb{R}^m$ is the image of *at least one* x in $\mathbb{R}^n$.

Definition. A mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is said to be __________ (or _________) if each b in $\mathbb{R}^m$ is the image of *at most one* x in $\mathbb{R}^n$.

Theorem. *Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then T is one-to-one (injective) if and only if the equation ________ has only the trivial solution.*

Theorem. *Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation, and let A be the standard matrix for T. Then:*

1. *T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ (is surjective) if and only if ______________________;*
2. *T is one-to-one (injective) if and only if ____________________________________.*

Example. Let T be the linear transformation whose standard matrix is

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

Does T map $\mathbb{R}^4$ onto $\mathbb{R}^3$ (is it surjective)? Is T a one-to-one mapping (injective)?

Since A happens to be in echelon form, we can see at once that A has a pivot position in each row. By Theorem 4 in Section 1.4 of the book (also in the previous notes), for each $b \in \mathbb{R}^3$, the equation $Ax = b$ is consistent. In other words, the linear transformation T maps $\mathbb{R}^4$ (its domain) onto (surjectively)$\mathbb{R}^3$. However, since the equation $Ax = b$ has a free variable (because there are four variables and only three basic variables), each b is the image of more than one x. That is, T is not one-to-one (injective).

Matrix Operations

Addition and Scalar Multiplication

In this section we talk about the possible operations we can do between matrices. We will denote a_{ij} the element of an $m \times n$ matrix A that is in the i-th row and the j-th column, with $i \leq m$ and $j \leq n$. Of course, the elements in the **diagonal** are a_{ii}. A **diagonal matrix** is a matrix whose non-diagonal entries are zero. The **zero matrix** is the matrix whose entries are all zero.

We will say two matrices are **equal** if they have the same size and their corresponding entries are equal. If A and B are both $m \times n$ matrices, then the **sum** $A + B$ is the sum of their columns, or equivalently, the sum of their corresponding entries. If $r \in \mathbb{R}$, then the **scalar multiple** rA is the matrix whose columns are multiplied by r, or equivalently, whose entries are multiplied by r.

Theorem. *Let A, B and C be matrices of the same size, and let r and s be scalars.*

1. $A + B =$ _____
2. $(A + B) + C =$ ___________
3. $A + 0 =$ _
4. $r(A + B) =$ _______
5. $(r + s)A =$ _______
6. $r(sA) =$ _____

Matrix Multiplication

Given linear transformations $S : \mathbb{R}^p \to \mathbb{R}^n$ and $T : \mathbb{R}^n \to \mathbb{R}^m$ we can form the composition $T \circ S$ (read as "T after S" or "T composite with S"), defined as

$$(T \circ S)(x) = \underline{\hspace{2cm}}$$

for $x \in \mathbb{R}^p$.

Exercise. Using the fact that S and T are linear transformations, show that their composition is also a linear transformation.

Write A for the standard matrix ($m \times n$) of T and write B for the standard matrix ($n \times p$) of S. If $B = [b_1 \cdots b_p]$ and $x = (x_1, \ldots, x_p)$, then computing $S(x)$ amounts to

$$Bx = \underline{\hspace{3cm}}.$$

If we now want to apply T to $S(x)$ to get $T(S(x))$, we multiply by A on the left and use its linearity,

$$A(Bx) = \underline{\hspace{12cm}}.$$

Therefore, we will define the **matrix product** AB of A and B to be the standard $(m \times p)$-matrix of $T \circ S$:

$$AB = \underline{\hspace{4cm}}.$$

Remark. Remember that the sizes of the matrices have to match appropriately in order to be able to multiply them. The columns of the first matrix has to match the rows of the second. The resulting size will have the numbers of rows of the first matrix and the number of columns of the second.

Exercise. Compute the matrix product AB where

$$A = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}, B = \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}.$$

Properties of Matrix Multiplication

Recall that I_m represents the $m \times m$ identity matrix (zero matrix with ones in the diagonal) and that $I_m x = x$ for all $x \in \mathbb{R}^m$.

Theorem. *Let A be an $m \times n$ matrix, and let B and C have sizes for which the indicated sums and products are defined.*

1. $A(BC) =$ ______
2. $A(B + C) =$ ________
3. $(B + C)A =$ ________
4. $r(AB) =$ _____________ *for any scalar r*
5. $I_m A =$ _______

Remark. <u>CAREFUL</u>: There are some properties that do not hold in general:

1. The commutative property: $AB \neq BA$
2. Cancellation laws: if $AB = AC$ we CANNOT say that $B = C$
3. If $AB = 0$ we CANNOT say that either $A = 0$ or $B = 0$

Definition. We define the powers of a matrix by $A^k = \underbrace{A \cdots A}_{k \text{ times}}$.

Definition. Given an $m \times n$ matrix A, the **transpose** of A is the $n \times m$ matrix A^T, whose columns are formed from the corresponding rows of A.

Example. Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, then $A^T = \left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right)$.

Theorem. *Let A and B denote matrices whose sizes are appropriate for the following sums and products.*

1. $(A^T)^T =$ _
2. $(A + B)^T =$ _______
3. *For any scalar r,* $(rA)^T =$ ____
4. $(AB)^T =$ _____